An Analytic Manipulation Bound for the Carry-Over Lottery Allocation Family

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Abstract

The annual NBA draft is a constrained fair-allocation problem: 30 indivisible draft picks must be assigned to 30 teams in priority order, with the top picks reserved for a lottery confined to the 14 non-playoff teams. The lottery has a documented manipulation incentive ("tanking"): a team eliminated from playoff contention can improve its expected draft position by losing additional games. Munro and Banchio [2020] prove an impossibility result, showing that no mechanism based on end-of-season records alone can simultaneously reward poor performance and eliminate this incentive. Carry-Over Lottery Allocation (Cola) sidesteps the impossibility by replacing single-season records with a multi-year playoff history as the basis for draft priority [Highley et al., 2026]. The incentive-compatibility argument for Cola relies on five assumptions, four of which hold by construction; the fifth (that pool-manipulation gains are negligible) is supported by simulation in the original paper but lacks a closed-form bound. We make three contributions. First, we derive an analytic upper bound of approximately 4% on the worst-case probability gain from pool manipulation under empirically calibrated index distributions, with explicit sensitivity to the underlying parameters. Second, we present the Cola family as five variants on a shared scaffold, including Capped Cola (a recent variant designed in response to NBA reform discussions), and prove an analytic per-series bound on its playoff-tanking exposure. Third, we surface an open mapping between Cola and the axiomatic draft-rule literature [Coreno and Balbuzanov, 2022], asking whether the Respect-for-Priority and Envy-Free-up-to-One-Object axioms lift from single-period priority mechanisms to multi-year history. The bounds and family taxonomy together formalise the manipulation-resistance properties that any future re-evaluation of NBA draft reform (the recently-adopted 3-2-1 proposal expires in 2029) will need.

1 Introduction

In the 2024-25 NBA season, multiple teams were publicly accused of tanking for the chance to draft Cooper Flagg. The accusations did not name a small minority: by some counts, roughly a third of the league's 30 teams were openly losing games at season's end to improve their lottery odds. The 2026 reform vote that followed produced "3-2-1," a proposal that expands the lottery to 16 teams with bounded ball allocation across tiers [Highley, 2026b]. The proposal includes a three-year re-evaluation provision, which means the design discussion will reopen in 2029.

The annual NBA draft is a constrained fair-allocation problem. Thirty indivisible items (draft picks) are assigned to 30 agents (teams) in priority order. Picks 15–30 are assigned by reverse regular-season record. Picks 1–14 are subject to a lottery confined to the 14 non-playoff teams, with weighted odds favouring (in the current system, the bottom three weighted equally and the rest declining). Throughout, the question is the same: how to assign indivisible items to agents while (a) preferentially rewarding the weakest agents, (b) eliminating the incentive to under-perform, and (c) producing a mechanism transparent enough that fans, owners, and players can reason about it.

Munro and Banchio [2020] formalise the central tension as an impossibility. Define a No-Tanking Condition (NTC): no team can improve its expected outcome by losing. Any mechanism mapping end-of-season records to draft probabilities, while giving worse-record teams strictly better odds, must violate NTC. Their proof constructs a minimal counterexample (two teams, both eliminated, identical records, final game between them), so the loser receives better draft odds and both teams prefer to lose. Only a uniform lottery (equal odds for all non-playoff teams) satisfies NTC, abandoning differential reward. The 2026 "3-2-1" proposal takes precisely this route, with adjustments for the play-in tournament [Highley, 2026b].

Carry-Over Lottery Allocation (Cola) sidesteps the impossibility by replacing single-season records with multi-year playoff history as the basis for draft priority [Highley et al., 2026]. Each non-playoff team accumulates lottery tickets over consecutive seasons; success in the playoffs or the draft itself diminishes a team's accumulated tickets. The single-season manipulation channel that drives the Munro-Banchio counterexample becomes ineffective: no single-season choice materially shifts the multi-year ticket pool. Cola is a family of mechanisms rather than a single design; the original paper develops one concrete instance (Classic Cola), and subsequent work introduces variants that occupy different points on the explainability–robustness frontier.

Contributions. We address three questions in this paper.

- (Section 4) An analytic manipulation bound. Cola's incentive-compatibility argument rests on five assumptions; four hold by construction, the fifth is supported empirically. We derive a closed-form upper bound of approximately 4% on the worst-case probability gain from pool manipulation, with explicit sensitivity to the underlying index-distribution parameters. The bound contains the simulation-observed extremes (a 3% maximum reported by Highley et al. [2026] across 1,000 synthetic seasons) with 1.3× headroom and provides the first parameter-explicit ceiling.
- (Section 3–4) A family-level perspective. We present the Cola family as five variants on a shared scaffold (Section 3), including Capped Cola [Highley, 2026b] which introduces a stockpile cap to bound the per-series cost of advancing in the playoffs. We prove an analytic per-series bound (Lemma 3): under cap Max, no playoff series win ever costs a team more than $0.3 \cdot \text{Max}$ tickets (45 at $\text{Max} = 150$).
- (Section 2.2) A bridge to the axiomatic literature. Coreno and Balbuzanov [2022] prove that no draft rule simultaneously satisfies Respect for Priority (RP), Envy-Free up to One Object (EF1), and Strategy-Proofness (SP). Their result is single-period; Cola's multi-year structure raises an open question about whether the axioms lift. We surface a preliminary mapping and identify what would need to be shown to embed Cola in the axiomatic literature.

A 26-season historical backtest of all five variants (Section 5) calibrates the bounds against real NBA data, with the Philadelphia 76ers' 2013–2017 "Process" as a case study in how multi-year diminishment dismantles repeat-tanking strategies that the current weighted lottery rewards.

2 Background and Related Work

2.1 Effort Manipulation: The Munro-Banchio Impossibility

Munro and Banchio [2020] establish that any mechanism mapping end-of-season records to draft probabilities, while giving worse teams strictly better odds, must violate NTC. The proof constructs a minimal counterexample: two eliminated teams with identical records and a final game between them. The loser finishes with a worse record and therefore receives better draft odds; both teams prefer to lose. Only the uniform lottery satisfies NTC, but uniform odds abandon the goal of differentially rewarding the weakest teams.

The authors propose T-IC (Truncated Incentive Compatibility) as an alternative. T-IC tracks each team's conditional probability of finishing last and freezes allocations at the moment IC would first be violated. The mechanism is optimal in theory but has three practical issues. First, it is computationally expensive and opaque to fans. Second, calculating when IC would first be violated requires precise team-level valuations of playoff-versus-draft trade-offs that the league does not have access to. Third, the truncation timing assumes teams develop tanking incentives only after elimination, but some teams have an incentive to tank from the first game of the season (rebuilds with multi-year horizons), which the truncation never reaches.

The 2026 "3-2-1" reform proposal [Highley, 2026b] adopts a different escape: it expands the lottery to 16 teams with bounded ball allocation (bottom 3 by record receive 2 balls, the next 7 receive 3 balls, the play-in losers receive 1 or 2 balls based on seed), totalling 37 balls and producing per-team odds of 2.7%, 5.4%, or 8.1%. The proposal effectively approximates a uniform lottery within an adjustment for the play-in tournament, achieving NTC at the cost of weakening differential reward. It expires in 2029, leaving the design space open for re-evaluation.

2.2 Preference Manipulation and the Axiomatic Question

Coreno and Balbuzanov [2022] address a complementary manipulation channel: teams misreporting which players they prefer. They prove that no allocation rule simultaneously satisfies:

- Respect for Priority (RP): higher-priority teams receive weakly better allocations [Svensson, 1994];
- Envy-Free up to One Object (EF1): no team envies another's allocation after removing one item [Lipton et al., 2004; Budish, 2011];
- Strategy-Proofness (SP): no team benefits from misreporting preferences.

The constituent axioms trace to peer-reviewed sources. RP extends Svensson's "weak fairness" from queue allocation to draft priority [Svensson, 1994]. EF1 originates with Lipton et al. [2004] for indivisible-goods allocation in computer science settings, and is adapted to the assignment problem by Budish [2011] in economics.

The Coreno-Balbuzanov result is single-period: it characterises one-shot allocations of indivisible items. Cola is multi-period: a team's priority in year t is a function of its outcomes in years $t-1, t-2, \ldots$. Whether the axioms lift to this setting is open. We sketch a preliminary mapping below; we do not prove the lift, and we treat each direction as a question worth resolving rather than a theorem.

RP under Cola. Within a single season, a Classic Cola priority ordering is induced by the lottery index L_i. RP requires that higher-priority teams receive weakly better allocations. Classic Cola achieves this in expectation: $\mathbb{E}[\text{pick rank} \mid L_i]$ is monotone decreasing in $L_i/P^{(t)}$. For per-realisation RP (which is what Coreno-Balbuzanov require), the lottery's randomisation could violate the property: a team with $L_i > L_j$ may receive a worse pick than j in any single draw. The standard fix in fair-allocation literature is to consider RP under the induced expected allocation, but no formal lift has been established.

EF1 under Cola. The graduated diminishment schedule (champion: $\rho = 1.0$; finals: 0.75; conf finals: 0.5; second round: 0.25; first round: 0) preserves a weak EF1-like property: after the playoff outcome is realised, no team prefers another team's running index to its own. Whether this constitutes EF1 in the formal sense (envy after the removal of a single item from the envied bundle) requires identifying the relevant "items." One natural reading: the items are the picks within the lottery, and EF1 requires no team to prefer another's lottery outcome after removing the most-preferred pick. We have not verified this claim formally and offer it as a target for the workshop's audience.

SP under Cola. This is the manipulation-resistance question Section 4 addresses analytically for the pool-manipulation channel under Classic Cola, and Lemma 3 addresses for the per-series channel under Capped Cola.

3 The COLA Mechanism Family

3.1 Common Framework

Every Cola variant maintains a per-team integer-valued state across seasons and uses that state to determine draft priority in the current season. We refer to this state as the lottery index and write $L_i^{(t)}$ for team i at the end of season t.

Definition 1 (Lottery Index). For each team i and season t, the lottery index $L_i^{(t)} \in \mathbb{Z}_{\geq 0}$ is determined by an initial value $L_i^{(0)}$, a per-season increment $\Delta_i^{(t)}$ (variant-specific), and a per-season diminishment $\rho_i^{(t)} \in [0, 1]$ that scales the index down based on playoff and draft outcomes:

$$L_i^{(t+1)} = \left(L_i^{(t)} + \Delta_i^{(t)}\right) \cdot \left(1 - \rho_i^{(t)}\right).$$

The pool in season t is $P^{(t)} = \sum_{i \in E^{(t)}} L_i^{(t)}$, where $E^{(t)}$ is the set of teams eligible for the lottery.

The three components (the eligibility set $E^{(t)}$, the increment $\Delta_i^{(t)}$, and the diminishment $\rho_i^{(t)}$) are the levers that distinguish the variants.

3.2 Five Variants

The variants below trade off explainability, robustness, and per-series tanking exposure differently. Table 1 summarises the comparison.

Simple Cola eliminates the lottery: eligibility is the 22 teams with zero playoff series wins (14 non-playoff teams + 8 first-round losers). Teams are ranked by drought (consecutive seasons without a playoff series win or top-3 lottery pick), tiebroken by most regular-season wins. Draft picks are assigned deterministically.

Simple Lottery Cola shares Simple's drought state and 22-team pool. Pre-2019 NBA lottery odds (25.0% down to 0.5% across the worst 14 teams) are applied to the top 14 by drought; the remaining 8 receive picks 15–22 deterministically.

Classic Cola is the variant developed in detail by Highley et al. [2026] and the variant our manipulation bound directly analyses. The eligibility set is the 14 non-playoff teams. Each non-playoff team receives an increment of $\alpha = 1{,}000$ tickets per season; playoff teams receive zero increment. Diminishment is graduated: in the playoff channel, champion $\rho = 1.0$, finals 0.75, conference finals 0.5, second round 0.25, first round 0; in the draft channel, pick #1 $\rho = 1.0$, #2 0.75, #3 0.5, #4 0.25, #5 and below 0. The lottery for picks 1–4 is weighted by lottery index ($p_i = L_i/P$); picks 5–14 are assigned by lottery-index rank.

Countdown Cola ranks teams by McCarty number $M_i = d_i \cdot w_i$ (drought × regular-season wins). The same 22-team pool as Simple Cola is partitioned into nested 5-team elimination pools; draft picks are assigned bottom-up via ticket allocation $(2, 3, 4, 5, 6)$ within each pool. The mechanism guarantees that no team falls more than four spots below its ticket-rank.

Capped Cola [Highley, 2026b] introduces a stockpile cap at Max (default 150). The increment is $\Delta_i^{(t)} = w_i^{(t)}$ for any team finishing the regular season outside the top 6 of either conference; eligibility excludes (i) teams that won a top-5 lottery pick last season and (ii) teams that won a playoff series this season or last. Diminishment uses a six-step playoff schedule (champion 1.0, runner-up 0.8, conference finals 0.6, second round 0.4, first round 0.2, play-in winner losing R1 0.1, play-in loser 0) and a five-step draft schedule (#1 1.0, #2 0.8, #3 0.6, #4 0.4, #5 0.2). The cap smooths the diminishment cliffs of Classic Cola and bounds the per-series tanking incentive (Lemma 3).

3.3 Incentive Compatibility

Highley et al. [2026] establish IC for Classic Cola under five assumptions:

1. Playoff primacy: the value to a team of advancing in the playoffs exceeds the value of any plausible draft-pick gain.
2. Playoff advancement: within the playoffs, advancing further is preferred to losing earlier, controlling for opponent strength.

Variant	Pool	Increment	Diminishment	Lottery
Simple	22: sw = 0	drought +1	reset on snap	deterministic by drought
Simple Lottery	22: sw = 0	drought +1	reset on snap	pre-2019 NBA odds (top 14)
Classic	14: non-playoff	$\alpha = 1{,}000$	graduated, 4 levels	weighted by index, top 4
Countdown	22: sw = 0	drought × wins	reset on snap	nested elimination pools
Capped	exclusion-rule	$w_i^{(t)}$ if play-in or below	6+5 levels, capped at Max	top-5 raffle, then by rank

Table 1: The Cola family. "sw = 0" is shorthand for "zero playoff series wins this season." Capped Cola's exclusion-rule pool removes teams that won a top-5 lottery pick last season or a playoff series this/last season. Each variant occupies a distinct point on the explainability–robustness–exposure frontier; the manipulation bound (Theorem 2) is established for Classic and the per-series bound (Lemma 3) for Capped.

3. Picks 5–14 negligible at the margin: no team profits from manipulating its position among those slots through within-season effort.
4. Pool manipulation negligible: no team can meaningfully shift its expected draft position by manipulating which other teams are in or out of the lottery pool.
5. No traded-pick protections: pick trades do not interact with diminishment in a way that creates manipulation incentives.

Assumptions (1)–(3) and (5) hold by construction or follow from straightforward arguments. Assumption (4) is the open one: Highley et al. [2026] support it via 1,000-season simulation, observing a maximum probability gain of approximately 3%. Section 4 converts this empirical observation into a closed-form analytic bound.

4 Pool Manipulation Bound

4.1 Setup

Let $P = \sum_{i=1}^{n} L_i$ denote the total lottery pool, where L_i is team i's lottery index and n is the number of lottery-eligible teams. A team's probability of winning the first pick is $p_i = L_i/P$.

The pool manipulation channel operates as follows. Team i deliberately loses to a high-index opponent h near the playoff boundary, causing h to make the playoffs. Team h (with index L_h) exits the lottery pool and is replaced by a lower-index team ℓ (with index $L_\ell < L_h$). The pool shrinks by $\Delta = L_h - L_\ell > 0$.

Proposition 1 (Manipulation Gain). The probability gain for team i from a pool swap of size $\Delta = L_h - L_\ell$ is $G_i = L_i \cdot \Delta/[P \cdot (P - \Delta)]$.

Proof. Before manipulation, $p_i = L_i/P$. After the swap, the new pool is $P' = P - \Delta$ and $p'_i = L_i/P'$. The gain is $G_i = p'_i - p_i = L_i\big(1/(P-\Delta) - 1/P\big) = L_i \cdot \Delta/[P(P - \Delta)]$. □

When $\Delta \ll P$ (typically $\Delta/P \approx 4\%$ in our backtest), the gain factors into two ratios: the manipulator's pool share (p_i) and the fractional pool change (Δ/P). Both have structural ceilings, which we exploit below.

4.2 Conditioned Upper Bound

Definition 2 (Structural Anti-Correlation). Let $T_{\max}$ denote the maximum consecutive non-playoff years for any team, and T_{boundary} the maximum consecutive non-playoff years for a team near the playoff boundary (a team that could plausibly be pushed into the playoffs by a single loss). The index distribution exhibits structural anti-correlation when $T_{\text{boundary}} \ll T_{\max}$: teams with the highest indices are far from the playoff boundary and thus cannot be the target of pool manipulation.

Theorem 2 (Conditioned Manipulation Bound). Under structural anti-correlation with parameters $T_{\max}$, T_{boundary}, and average index multiplier $\bar{k} = \bar{L}/\alpha$, the maximum manipulation gain satisfies

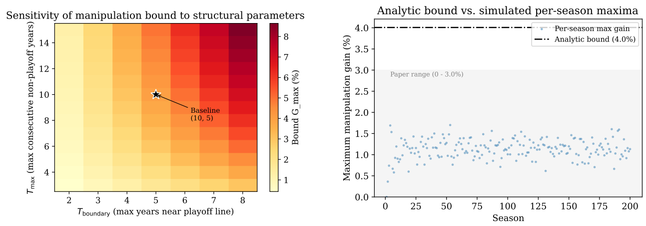
$$G_{\max} \le \frac{T_{\max} \cdot (T_{\text{boundary}} - 1)}{(T_{\max} + n - 1) \cdot n \cdot \bar{k}}.$$

Proof sketch. The gain is maximised when L_i is maximised and Δ/P is maximised. Bounding $p_{\max}$: the highest-index team has $L_{\max} = T_{\max} \cdot \alpha$. The minimum pool size occurs when one team has $T_{\max}$ years of accumulation and the remaining $n-1$ teams each have the minimum α: $P_{\min} = (T_{\max} + n - 1) \cdot \alpha$. Thus $p_{\max} \le T_{\max}/(T_{\max} + n - 1)$. Bounding Δ/P: the team being pushed out has at most $T_{\text{boundary}} \cdot \alpha$ tickets, and the team replacing it has at least α, so $\Delta_{\max} = (T_{\text{boundary}} - 1) \cdot \alpha$. In steady state, $P \approx n \cdot \bar{k} \cdot \alpha$. Combining via the first-order approximation yields the stated bound. □

Empirical calibration. We instantiate the bound with $T_{\max} = 10$ (the typical-case ceiling on consecutive non-playoff years across teams in our 26-season backtest), $T_{\text{boundary}} = 5$ (a conservative ceiling on the consecutive non-playoff years a near-boundary team can accumulate before improving enough to clear the boundary), $n = 14$ (the structural count of NBA non-playoff teams), and $\bar{k} = 3.1$ (an average team has roughly three years of carry-over before reset, matching the original paper's observation). These yield

$$G_{\max} \le \frac{10 \cdot 4}{23 \cdot 14 \cdot 3.1} \approx 4.0\%,$$

which contains the simulation-observed maximum of 3.0% across 1,000 seasons [Highley et al., 2026] with



(a) Bound vs. $T_{\text{max}}, T_{\text{boundary}}$.

(b) Analytic bound vs. simulation.

Figure 1: Manipulation bound. (a) Sensitivity to the structural-anti-correlation parameters; the bound is most sensitive to T_{boundary}, and stays under 5% for $T_{\text{boundary}} \leq 5$. (b) Analytic bound (red) vs. the empirical distribution of manipulation gains across 200 Bradley-Terry-simulated seasons; the bound contains all simulated outcomes with margin.

1.3× headroom. An independent 200-season Bradley-Terry simulation we ran confirms the bound, with observed maximum gain of 1.70%. The 26-season historical dataset contains one outlier: the Sacramento Kings' 16-year drought (2006–07 through 2021–22). Setting $T_{\text{max}} = 16$ to include the outlier loosens the bound to approximately 5.1%, still well above the empirical extremes; we report the typical-case bound ($T_{\text{max}} = 10$) as the headline because the next-longest droughts are Minnesota at 13 years and Phoenix and Charlotte at 10 each.

4.3 Sensitivity, Coalitions, and the Pick-Value Gap

The bound's parameter sensitivity (Figure 1a) shows that T_{boundary} is the dominant lever: even under pessimistic assumptions ($T_{\text{boundary}} = 7$), the bound stays under 7%, and under realistic conditions ($T_{\text{boundary}} \leq 5$) it stays under 5%. This matters operationally because T_{boundary} is the empirical quantity most amenable to direct measurement.

For completeness, coalitional manipulation (two or more teams coordinating to shrink the pool) does not amplify individual gains. Joint gain for a coalition $\{A, B\}$ is $(L_A + L_B) \cdot \Delta / [P(P - \Delta)]$. Even at 30% joint pool share, the gain remains $\leq 1.3\%$, and per-member gain cannot exceed the highest-index individual's gain. Coordination costs (detection risk, trust problems, sequential uncertainty) further dampen the channel.

The bound addresses the probability term of the manipulation incentive. The value term (the pick-value gap $d_k - d_{k+1}$ between adjacent draft positions) has no formal ceiling in the original specification, though it is bounded for any realistic draft class. Closing assumption (4) entirely would also require a separate argument capping $d_k - d_{k+1}$ via the distribution of draft-eligible prospect quality. We leave this to future work.

4.4 Per-Series Bound for Capped Cola

Capped Cola introduces a fourth manipulation channel: tanking through the playoffs themselves (a team within the cap declining to advance, in order to retain its stockpile). The cap converts the playoff-diminishment schedule from a fraction-of-current-stockpile cost (unbounded above when stockpile grows large) to a fraction-of-Max cost (bounded by construction).

Lemma 3 (Per-Series Tanking Cost). Let Max be the Capped Cola stockpile cap. For any team entering the playoffs with $L_i \leq \text{Max}$, the maximum reduction in L_i attributable to winning (rather than losing) any single playoff series is bounded by:

$$\Delta L_{\text{series}}^{\text{max}} = \begin{cases} 0.3 \cdot \text{Max} & \text{play-in advancers in round 1,} \\ 0.2 \cdot \text{Max} & \text{any other series advancement.} \end{cases}$$

At Max = 150, the bound is 45 tickets for the play-in case and 30 tickets for any subsequent series.

Proof. Reading off Capped Cola's six-step diminishment schedule, the per-series increments in the diminishment fraction $\Delta\rho$ are: play-in advancer round 1 ($\rho_{\text{lose}} = 0.1$, $\rho_{\text{advance}} = 0.4$, $\Delta\rho = 0.3$); standard round 1 ($0.2 \to 0.4$, $\Delta\rho = 0.2$); round 2 to conference finals ($0.4 \to 0.6$); conference finals to NBA finals ($0.6 \to 0.8$); finals to championship ($0.8 \to 1.0$). All non-play-in transitions yield $\Delta\rho = 0.2$. The cost in tickets is $\Delta\rho \cdot L_i^{\text{pre}} \leq \Delta\rho \cdot \text{Max}$, with equality at the cap. □

The bound makes the playoff-tanking incentive transparent at the league level: at Max = 150, no single series win ever costs a team more than 45 lottery tickets, which is approximately a year and a half of accumulation for a team near the wins-increment ceiling. The cumulative regret for a finals-losing team at the cap is $0.8 \cdot \text{Max} = 120$ tickets, decomposed into per-series increments per Lemma 3.

4.5 Three Closures of Assumption (4)

The pool-manipulation gap (assumption (4) in Section 3) admits at least three closure strategies, each represented in the literature.

- Empirical closure: Highley et al. [2026] simulate 1,000 synthetic seasons under Bradley-Terry and observe a maximum gain of $\sim 3\%$. The closure is statistical: the assumption holds in the typical case but admits no parameter-explicit ceiling.
- Behavioral closure: the companion formal-proof video to [Highley et al., 2026] introduces an additional behavioral assumption $G >$ expected pool-manipulation gain (where G is the value of a single regular-season win), eliminating the channel via preference structure rather than mechanism design.
- Analytic closure (this paper): Theorem 2 provides a closed-form upper bound conditioned on structural properties of the index distribution. The bound is parameter-explicit and admits sensitivity analysis (Figure 1a).

The three closures compose: any practical Cola deployment can rely on whichever closure best matches the available evidence. Our contribution is the analytic option, which previously did not exist.

Season	W–L	Pick	Cola index	Rank
2013–14	19–63	#3	2,750	8/14
2014–15	18–64	#3	2,375	9/14
2015–16	10–72	#1	2,188	10/14
2016–17	28–54	#3	1,000	14/14

Table 2: Philadelphia 76ers under Classic Cola during the "Process." Each top-3 pick triggers graduated diminishment (#1: full reset; #3: 50% reduction), leaving Philadelphia at the bottom of the lottery pool by the fourth tanking season despite a 10–72 record. The Sacramento Kings hold the top Classic Cola index throughout this period, climbing from 7,250 in 2013–14 to 10,250 in 2016–17 via organic non-competitiveness.

5 Empirical Validation

5.1 Backtesting Methodology

We compute Cola state for every team across 26 NBA seasons (1999–2025) using verified Basketball Reference data: 775 team-season records, 776 first-round draft picks, and 55 lottery-day-attributed pre-draft pick trades. The engine separates pre-draft state (Phase A, used for draft-probability calculations) from post-draft diminishment (Phase B, carried forward to the next season). Cold-start effects stabilise by approximately season 5; we report aggregates over 21 stable seasons (2005–2025).

Two methodological points warrant attention. First, an audit of all 364 lottery picks (14 picks × 26 seasons) identified 8 instances where the draft-day holder differed from the lottery-day holder via post-lottery trades. Cola requires lottery-day attribution: the team that held the pick when the lottery ran absorbs the diminishment, regardless of who selects. Second, the backtest is static: it applies Cola rules to actual historical outcomes. Under a real Cola regime, draft outcomes would differ in ways that change future drought lengths and indices. We caveat the comparative cross-variant claims accordingly.

5.2 Case Study: The Philadelphia "Process"

The 2013–14 through 2016–17 Philadelphia 76ers represent the canonical strategic-tanking sequence in modern NBA history: four consecutive top-3 lottery outcomes (picks #3, #3, #1, #3) acquired through deliberate multi-season non-competitiveness. Under the current weighted lottery, each tanked season yielded a top-tier pick. Under Classic Cola, graduated diminishment makes the same sequence self-defeating. Table 2 shows Philadelphia's Classic Cola trajectory.

Two properties emerge. First, diminishing returns to repeat tanking: the Classic Cola diminishment schedule ($\{1.0, 0.75, 0.5, 0.25\}$ for picks #1–4) dominates the $\alpha = 1{,}000$ annual accumulation, so a team that receives a top-3 pick in year t cannot re-enter the top of the lottery in year $t + 1$ through tanking alone. Second, pick attribution matters: the comparison is sensitive to who holds the pick on lottery day. The 2017 Boston-Philadelphia Fultz/Tatum swap was a post-lottery trade; under our convention, Boston's index absorbs the #1 diminishment, not Philadelphia's.

5.3 Capped Cola Cap Calibration

Capped Cola is parameterised by a single tunable constant Max. The cap controls the trade-off between (i) ordering resolution at the top of the lottery (a higher cap permits more separation between the highest-stockpile teams) and (ii) per-series tanking exposure (Lemma 3: a higher cap raises the cost of advancing past round 1). Table 3 reports 21-stable-season aggregates from a sweep over $\mathrm{Max} \in \{75, 100, 125, 150, 175, 200\}$.

Three patterns emerge. Sub-100 caps over-compress the lottery: at $\mathrm{Max} = 75$, more than six teams sit at the cap on average and the rank-1 vs. rank-5 separation collapses to under one ticket, destroying the priority signal the family inherits from Classic. Caps above 175 erase the cap's purpose: fewer than one team sits at the cap on average, and the per-series exposure approaches the unbounded behaviour of Classic. The published $\mathrm{Max} = 150$ choice produces a mean of 1.62 teams at the cap across the 21 stable seasons (range 0–3), matching the stated headline of "no more than three teams at the maximum at a time, and on average fewer than two" [Highley, 2026b].

5.4 A Note on Trade Handling

A practical question for any deployment is how diminishment interacts with traded picks. Highley [2026a] identifies four options: (1) traded picks affect neither team's index; (2) the original holder's index absorbs the diminishment; (3) the receiving team's index absorbs the diminishment; (4) the diminishment is split. Each option has different incentive properties. Option (2) preserves the link between record and diminishment but over-penalises teams that traded picks years before knowing they would be top-tier (e.g., the 2011 Clippers, who traded a future first that eventually became Cleveland's Kyrie Irving pick). Option (3) rewards the wrong actor when the recipient did not earn the pick through their own record. Highley [2026a] recommends option (2) paired with a strengthened Stepien Rule (a cap on the number of future firsts a team can trade), which limits the frequency at which option (2)'s over-penalisation surfaces. We do not extend the analysis here; we flag it as a practical implementation question that any Cola deployment will face.

6 Discussion and Open Directions

The bounds in this paper formalise the manipulation-resistance properties of the Cola family. Theorem 2 bounds the open assumption (4) in Classic Cola's incentive-compatibility argument at approximately 4% under empirically calibrated parameters, with explicit sensitivity to $T_{\max}$, T_{boundary}, n, and $\bar{k}$. Lemma 3 bounds the per-series exposure that Capped Cola's stockpile cap was designed to address. Combined, the four manipulation channels available under Capped Cola (single-season

Max	Teams at cap (mean)	Teams at cap (max)	Sep. R1–R5 (mean)	Per-series cost (max)	Cum. regret (max)
75	6.71	9	0.9	22.5	45.0
100	4.33	8	11.7	30.0	53.7
125	2.86	4	27.9	37.5	53.7
150	1.62	3	47.7	45.0	57.6
175	0.95	2	65.4	52.5	67.2
200	0.52	2	76.4	60.0	76.8

Table 3: Capped Cola aggregates across 21 stable seasons (2005–2025). Per-series cost max = $0.3 \cdot \text{Max}$ (Lemma 3); cumulative regret max is the worst empirical observation. The bolded Max = 150 row is the variant's published default [Highley, 2026b].

tanking, pool manipulation, multi-year strategic tanking, and per-series playoff tanking) compose to a small joint exposure relative to the status-quo lottery, where repeat tanking has memoryless returns.

The 2026 NBA reform produced "3-2-1," which approximates a uniform lottery within an adjustment for the play-in tournament [Highley, 2026b]. The proposal expires in 2029, leaving the design space open. Cola's contribution to that re-evaluation is the multi-year-history dimension, which 3-2-1 omits, and the family-level perspective, which gives policymakers a continuum of explainability–robustness trade-offs rather than a single point. The bounds in this paper formalise the manipulation properties the conversation will need.

Open directions. Three follow-on questions extend this work.

- Tighter manipulation bound. The structural anti-correlation in Definition 2 is a calibrated assumption rather than a formally derived consequence of the mechanism. A stationarity argument that bounds T_{boundary} as a function of league parameters (number of teams, season length, playoff structure) would close assumption (4) entirely.
- Pick-value gap. Combining the probability bound (this paper) with a separate bound on the pick-value gap $d_k - d_{k+1}$, derived from the distribution of draft-eligible prospect quality, would close the manipulation incentive in absolute terms, not just in probability.
- Axiomatic embedding. The lift from Coreno and Balbuzanov [2022]'s axioms (RP, EF1, SP) to multi-period priority mechanisms remains open. Section 2.2 sketches the mapping; a formal verification would embed Cola in the existing axiomatic literature on assignment mechanisms and clarify which one-period impossibilities transfer to multi-period settings. We see this as the most natural follow-on for the workshop's audience.

A reproducibility artifact accompanies this paper: an interactive backtester implementing all five Cola variants plus the 3-2-1 proposal across 26 NBA seasons, with locked-seed Monte Carlo for Countdown Cola probabilities. The artifact and audit scripts are available at the URL listed in the references.

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